

17.4

Ingress & TLS Termination

HTTP routing, cert-manager, Nginx Ingress

Java Backend Architecture Interview Series • Tutorial Edition • Easy Diagrams & Examples

Why Ingress?

A LoadBalancer Service gives you one external IP per service. If you have 20 microservices, you'd need 20 load balancers — expensive! An Ingress is a single entry point (one load balancer) that routes different URLs to different services based on hostname or path.

■ **Simple Analogy:** A hotel lobby (Ingress) has one entrance. Staff (routing rules) direct guests: guests going to the spa go left, restaurant guests go right, rooms guests take the elevator. One entrance, many destinations.

Diagram — Ingress Routing Flow

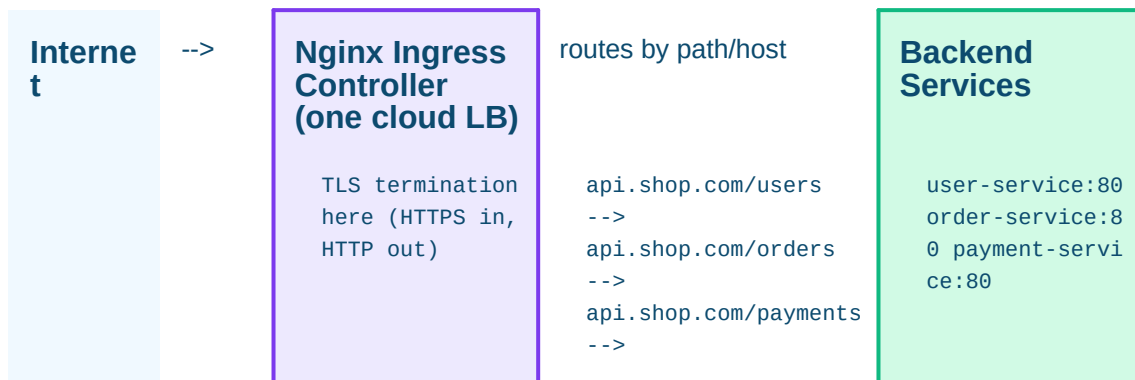


Figure 4: One Ingress handles all external traffic with TLS, fans out to internal services

Ingress YAML — Path-Based Routing

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: api-ingress
  annotations:
    nginx.ingress.kubernetes.io/rewrite-target: / # strip path prefix
spec:
  ingressClassName: nginx
  tls:
  - hosts: [api.shop.com]
    secretName: api-tls-cert # TLS cert stored as K8s Secret
  rules:
  - host: api.shop.com
    http:
      paths:
      - path: /users
        pathType: Prefix
        backend:
          service: { name: user-service, port: { number: 80 } }
      - path: /orders
        pathType: Prefix
        backend:
          service: { name: order-service, port: { number: 80 } }
```

Automatic TLS with cert-manager

cert-manager is a K8s add-on that automatically obtains TLS certificates from Let's Encrypt (free!) and renews them before expiry. You just add one annotation to your Ingress.

```
# Add to Ingress metadata to auto-provision a Let's Encrypt cert:
annotations:
cert-manager.io/cluster-issuer: letsencrypt-prod

# cert-manager flow:
# 1. Sees the annotation
# 2. Calls Let's Encrypt ACME API
# 3. Proves domain ownership (HTTP-01 or DNS-01 challenge)
# 4. Stores cert in Secret 'api-tls-cert'
# 5. Auto-renews 30 days before expiry
```

■ **Real-World Example:** Shopify runs 1000s of merchant storefronts on Kubernetes. A single Nginx Ingress routes traffic to merchant-specific services by hostname (merchant123.myshopify.com). cert-manager manages TLS for all custom domains automatically.